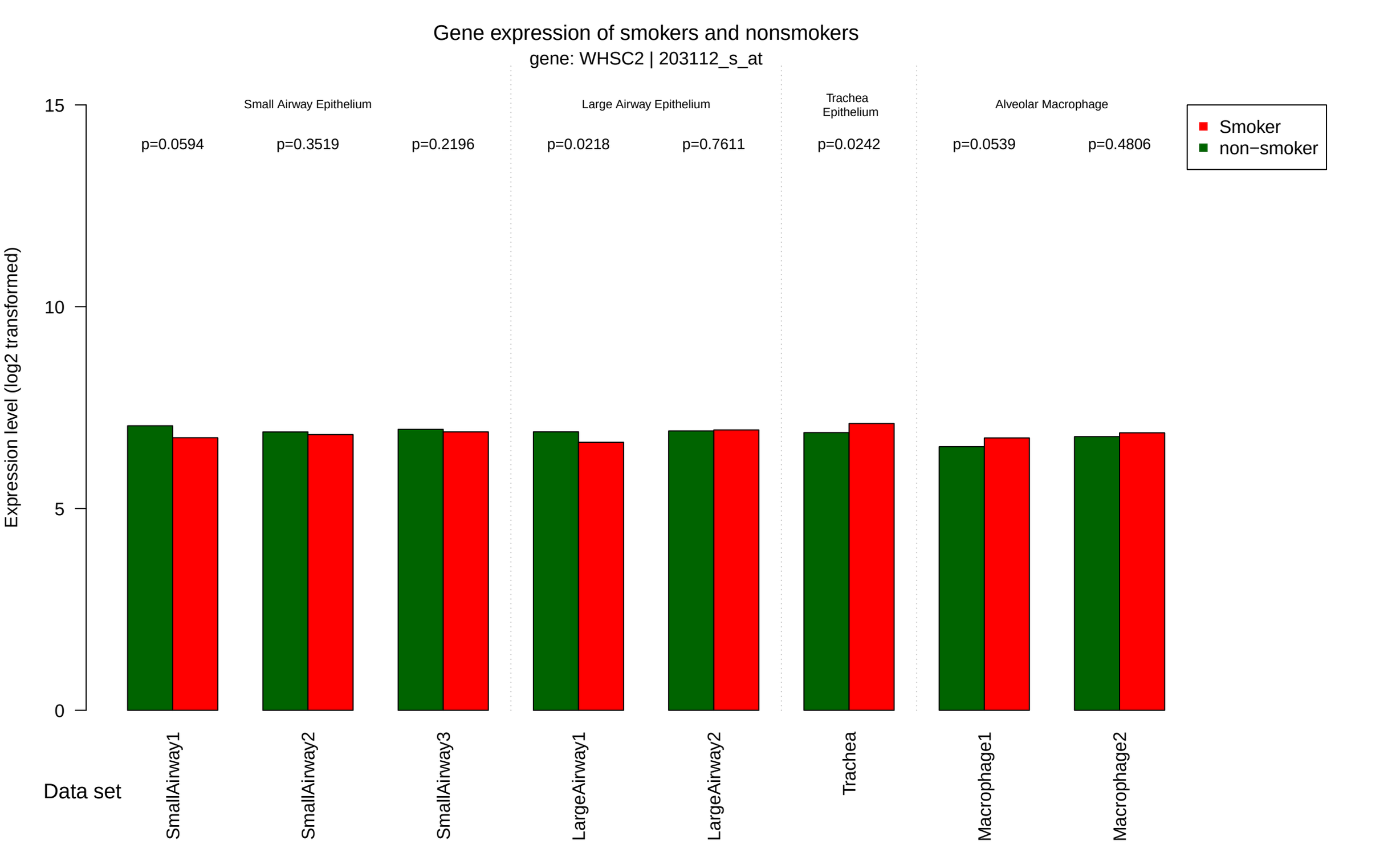
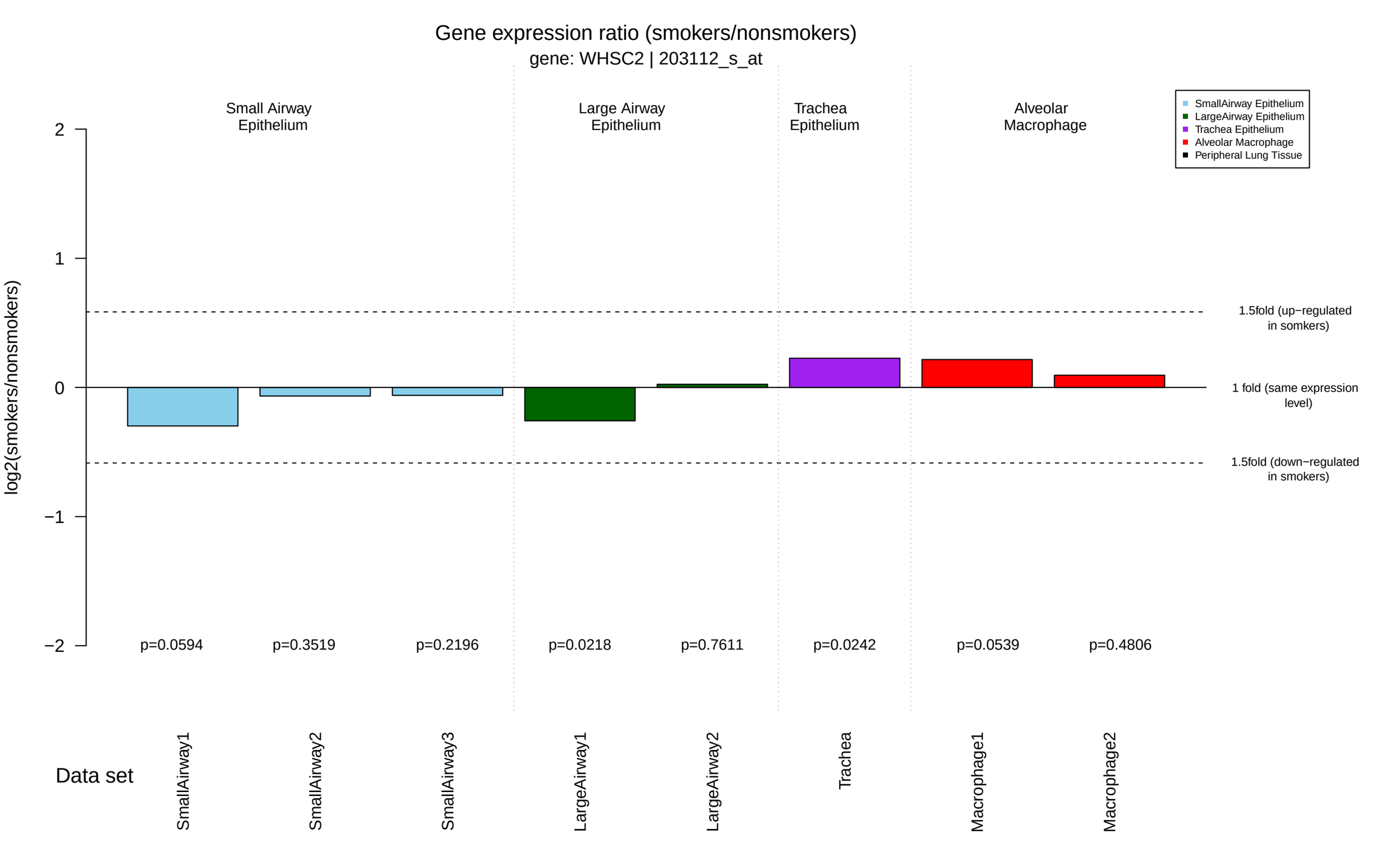
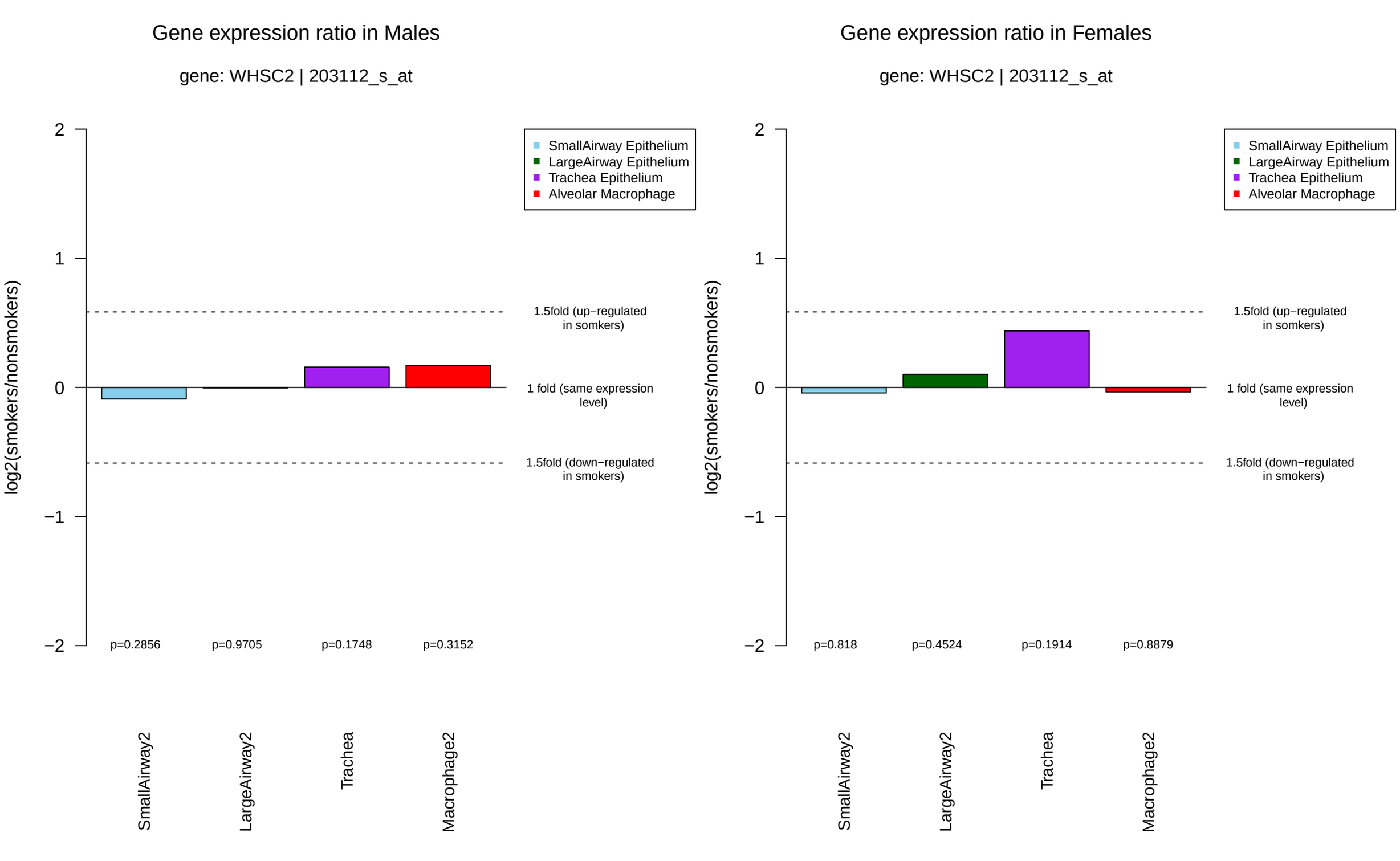
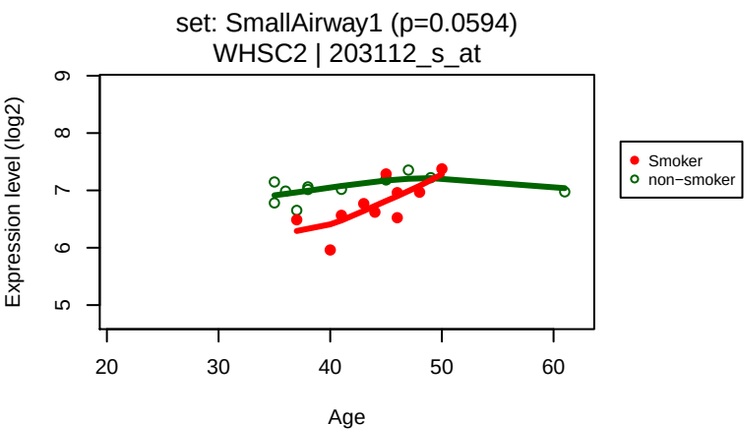
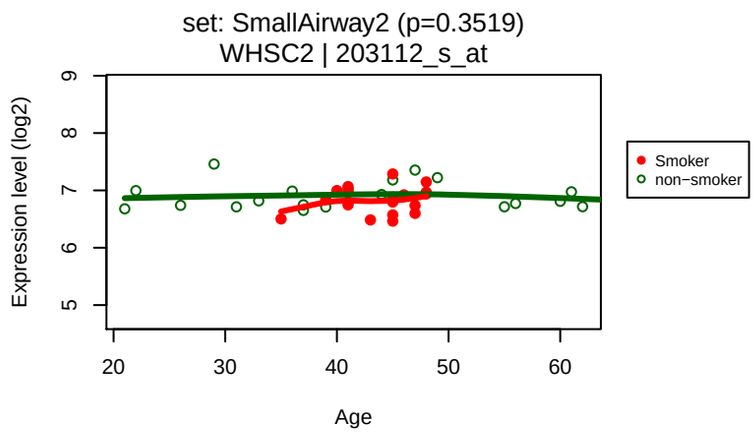
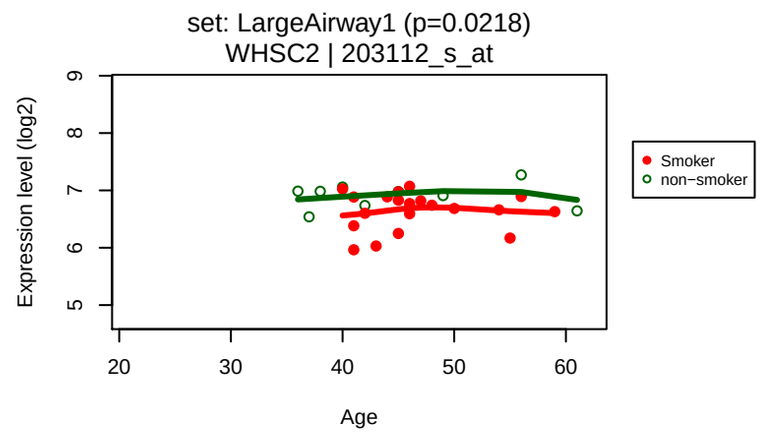

Gene expression of smokers and nonsmokers
gene: WHSC2 | 203112_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0594
p=0.3519
p=0.2196
p=0.0218
p=0.7611
p=0.0242
p=0.0539
p=0.4806
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

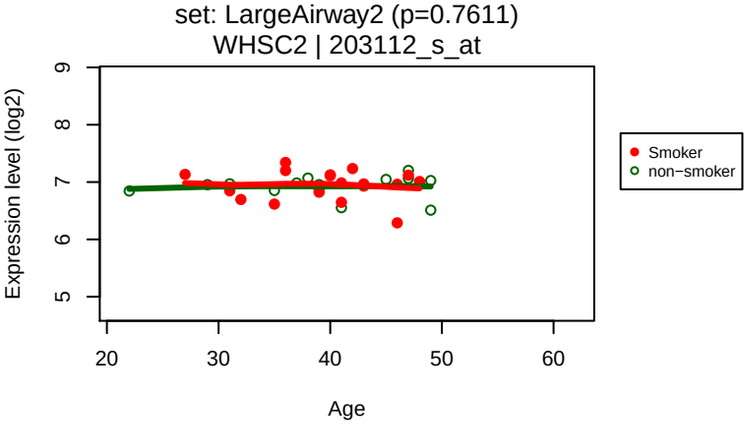

Gene expression ratio (smokers/nonsmokers)
gene: WHSC2 | 203112_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0594
p=0.3519
p=0.2196
p=0.0218
p=0.7611
p=0.0242
p=0.0539
p=0.4806
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

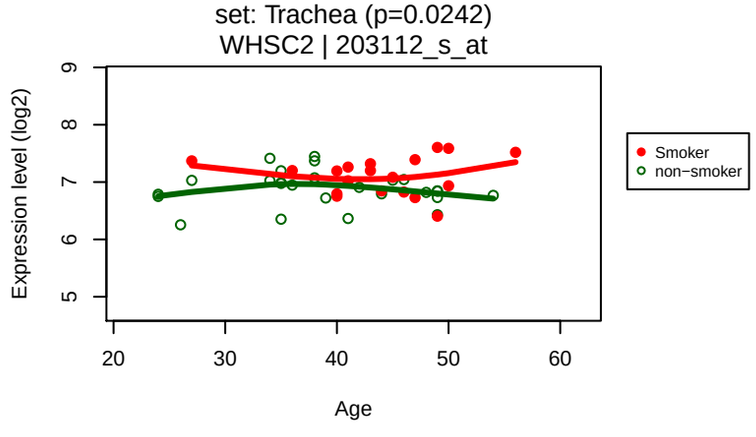

Gene expression ratio in Males
gene: WHSC2 | 203112_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2856
p=0.9705
p=0.1748
p=0.3152
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: WHSC2 | 203112_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.818
p=0.4524
p=0.1914
p=0.8879
SmallAirway2
LargeAirway2
Trachea
Macrophage2

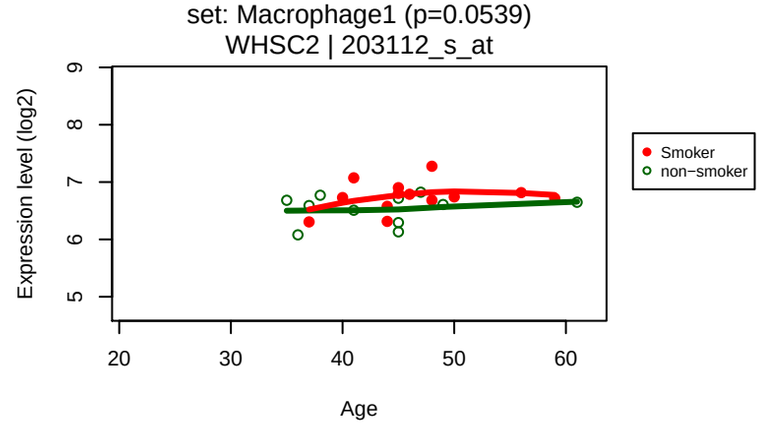

set: SmallAirway1 (p=0.0594)
WHSC2 | 203112_s_at
Expression level (log2)
Age
Smoker
non−smoker

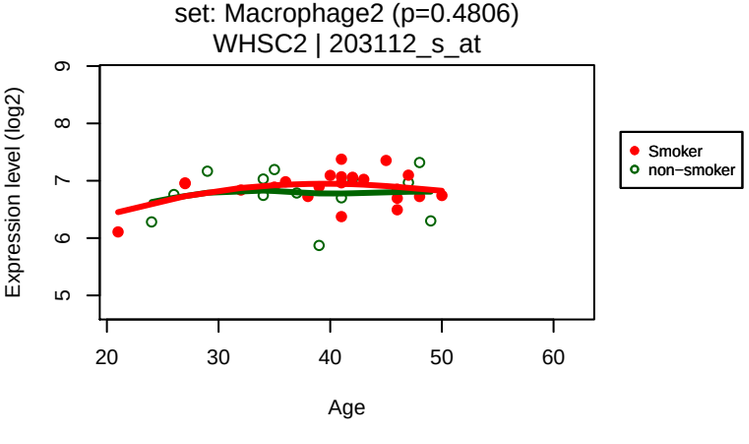

set: SmallAirway2 (p=0.3519)
WHSC2 | 203112_s_at
Expression level (log2)
Age
Smoker
non−smoker

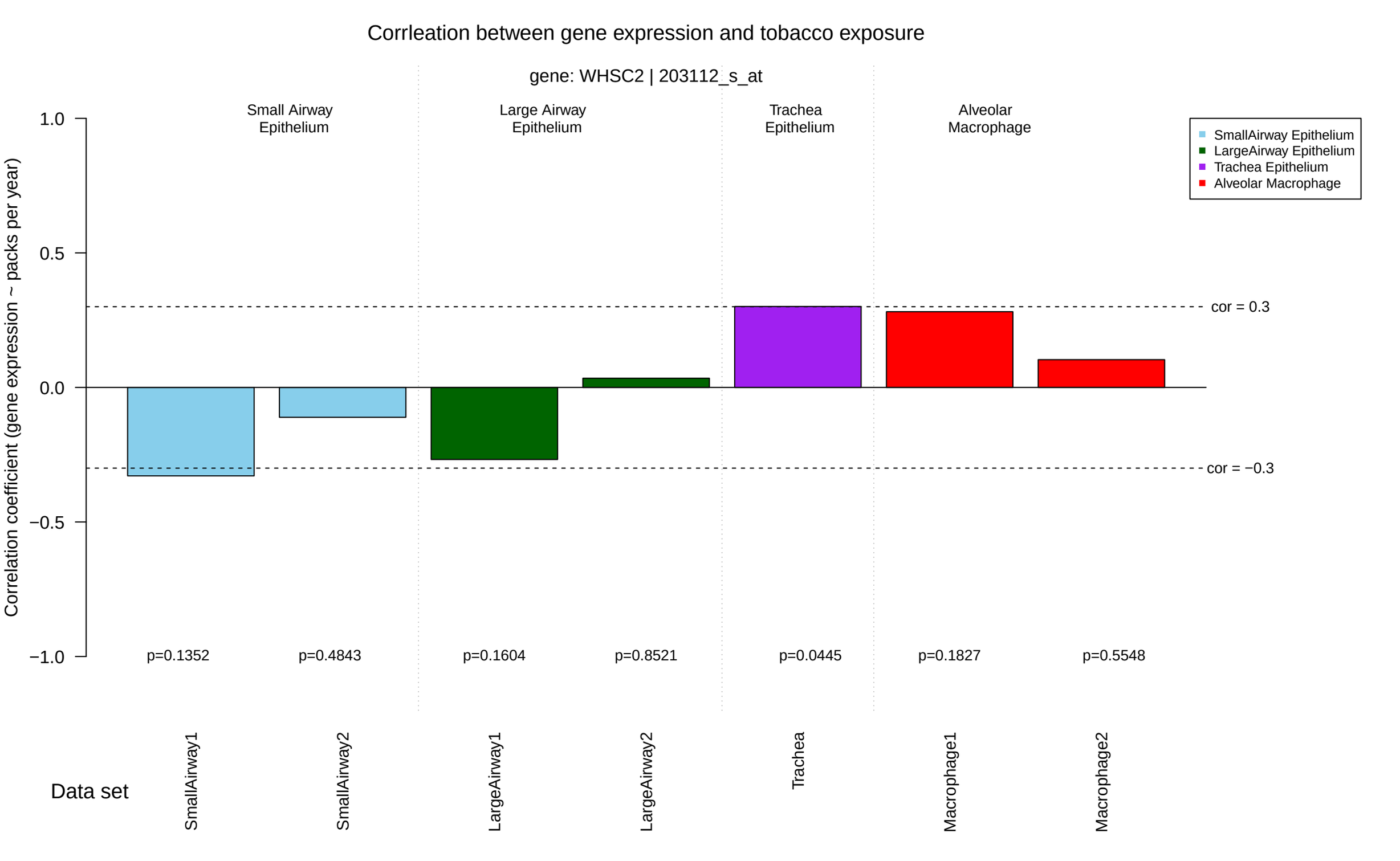

set: LargeAirway1 (p=0.0218)
WHSC2 | 203112_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7611)
WHSC2 | 203112_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0242)
WHSC2 | 203112_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0539)
WHSC2 | 203112_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4806)
WHSC2 | 203112_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: WHSC2 | 203112_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1352
p=0.4843
p=0.1604
p=0.8521
p=0.0445
p=0.1827
p=0.5548
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2